

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	A	B	B	D	C	C	B	B	A
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	A	C	A	A	C	A	B	D	C
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	31	6	7	1	6	10	2	5	7	5

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	A	B	C	D	A	D	C	D	A
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	C	C	A	C	A	D	D	C	A
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	5	8	68	1	8	2	3	4448	3663	6619

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	C	D	B	C	A	C	A	D	A
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	C	C	C	D	D	C	D	C	D
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	6	0	1	20	4	624	11	2	2	2

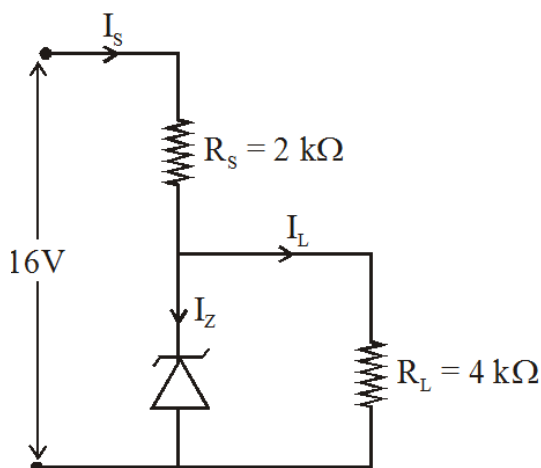
HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. Ans (B)

Maximum current will flow from zener if input voltage is maximum.



When zener diode works in breakdown state, voltage across the zener will remain same.

$$\therefore V_{\text{across } 4k\Omega} = 6V$$

$$\therefore \text{Current through } 4K\Omega = \frac{6}{4000} A = \frac{6}{4} \text{ mA}$$

Since input voltage = 16V

$$\therefore \text{Potential difference across } 2K\Omega = 10V$$

$$\therefore \text{Current through } 2k\Omega = \frac{10}{2000} = 5\text{mA}$$

$\therefore$ Current through zener diode

$$= (I_S - I_L) = 3.5 \text{ mA}$$

2. Ans (A)

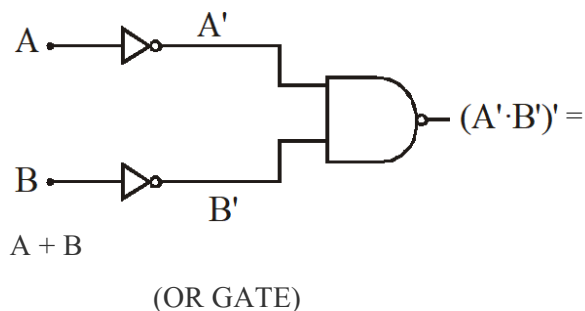
Method 1

Truth table can be formed as

A	B	Equivalent
0	0	0
0	1	1
1	0	1
1	1	1

Hence the Equivalent is "OR" gate.

Method 2



3. Ans (B)

Since the carrier frequency is distributed as band width frequency, so

$$10\% \text{ of } 10 \text{ GHz} = n \times 5 \text{ kHz}$$

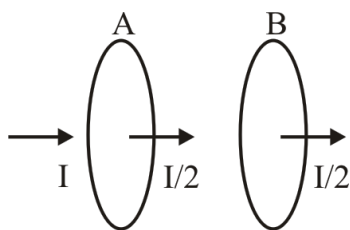
where n = no of channels

$$\frac{10}{100} \times 10 \times 10^9 = n \times 5 \times 10^3$$

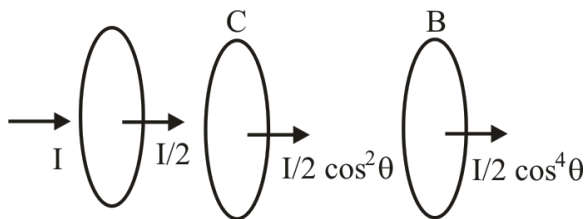
$$n = 2 \times 10^5 \text{ telephonic channels}$$

4. Ans (B)

Axis of transmission of A & B are parallel.



Now,

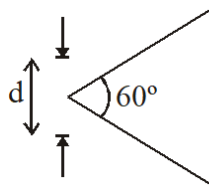


$$\frac{I}{2} \cos^4 \theta = \frac{I}{8} \Rightarrow \cos^4 \theta = \frac{1}{4}$$

$$\cos \theta = \frac{1}{\sqrt{2}}$$

$$\theta = 45^\circ$$

5. Ans (D)



In diffraction

$$d \sin 30^\circ = \lambda$$

$$\lambda = \frac{d}{2}$$

Young's fringe width

[d' – separation between two slits]

$$\beta = \frac{\lambda \times D}{d'}$$

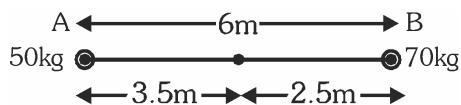
$$10^{-2} = \frac{d}{2} \times \frac{50 \times 10^{-2}}{d'}$$

$$10^{-2} = \frac{10^{-6} \times 50 \times 10^{-2}}{2 \times d'}$$

$$d' = 25 \mu\text{m}$$

6. Ans (C)

They will meet on their centre of mass



7. Ans (C)

$$Y = \frac{\left(\frac{F}{A}\right)}{\frac{\Delta \ell_1}{L}} \quad \dots(i)$$

$$Y = \frac{\left(\frac{F}{4A}\right)}{\frac{\Delta \ell_2}{2L}} \quad \dots(ii)$$

$$\frac{\Delta \ell_1}{\Delta \ell_2} = 2$$

8. Ans (B)

$$\tan \theta = \frac{(10 + 15) - (20)}{10} = \frac{5}{10} = \frac{1}{2} = \frac{a}{g}$$

$$\Rightarrow a = 5 \text{ m/s}^2$$

9. Ans (B)

$$\frac{1}{2} m \frac{4gR}{3} - \frac{mgR^2}{R} = -\frac{mgR^2}{(R + h_{\max})} \Rightarrow h_{\max} = 2R$$

$$\frac{1}{2} m \frac{4gR}{3} - \frac{mgR^2}{R} = \frac{1}{2} mu^2 - \frac{mgR^2}{2R}$$

$$\Rightarrow \left(-\frac{R}{3} + \frac{R}{2}\right) g = \frac{u^2}{2} \Rightarrow u = \sqrt{\frac{gR}{3}}$$

10. **Ans (A)**

For $A \rightarrow B$, $V \propto T \Rightarrow P = \text{const}$

$$\Rightarrow \Delta W_{AB} = nR\Delta T$$

$$= 2 \times R \times (750 - 250) = 1000R$$

For $B \rightarrow C$, $V = \text{const} \Rightarrow W_{BC} = 0$

For $C \rightarrow D$, $T = \text{const}$

$$\Rightarrow W_{CD} = nRT_C \ln\left(\frac{V_D}{V_C}\right)$$

$$\Rightarrow W_{CD} = 2 \times R \times 1000 \ln\left(\frac{20}{15}\right) = 2000R \ln\left(\frac{4}{3}\right)$$

Total work done

$$= \Delta W_{AB} + \Delta W_{BC} + \Delta W_{CD}$$

$$= 1000R + 0 + 2000R \ln \frac{4}{3}$$

$$= 2000R \left[\frac{1}{2} + \ln \frac{4}{3} \right]$$

11. **Ans (D)**

Equation of trajectory, $y = ax - bx^2$

For angle of projection, $\left. \frac{dy}{dx} \right|_{x=0, y=0} = a - 2bx$

$$\therefore \theta = \tan^{-1}(A)$$

For max. height,

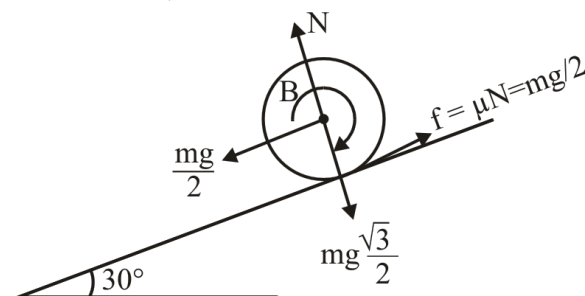
$$\frac{dy}{dt} = 0 \Rightarrow a \frac{dx}{dt} - 2bx \frac{dx}{dt} = 0$$

$$\Rightarrow x = \frac{a}{2b}$$

$$y = ax - bx^2 = \frac{a^2}{4b}$$

12. **Ans (A)**

Here kinetic friction force will balance the force of gravity along the incline. So it will rotate at its initial position and will not move up or down. As its angular velocity becomes zero (friction also becomes zero), it will move downwards.



13. **Ans (C)**

$$A = A_0 e^{-kt}$$

$$\frac{A_0}{4} = A_0 e^{-k(1)}$$

$$\frac{1}{4} = e^{-k}$$

$$A = A_0 e^{-k(3)}$$

$$A = A_0 (e^{-k})^3$$

$$A = \frac{A_0}{4^3}$$

$$x = 4^3$$

14. **Ans (A)**

For A, time period $T_A = 16$ s (distance between two adjacent crests)

for B, time period $T_B = 2(20 - 8) = 24$ s (length

between the crest and trough shown 20 s $- 8$ s $= 12$ s

Also, amplitudes $a_A = 10$ cm, $a_B = 5$ cm

$$\text{Now, } \frac{(V_{\max})_A}{(V_{\max})_B} = \frac{\omega_A a_A}{\omega_B a_B}$$

$$\frac{\left(\frac{2\pi}{T_A}\right) a_A}{\left(\frac{2\pi}{T_B}\right) a_B} = \frac{T_B a_A}{T_A a_B} = \frac{24 \times 10}{16 \times 5} = \frac{3}{1}$$

15. **Ans (A)**

Distance between two position of bridge is $\lambda/2$ (Distance between two nodes)

$$\frac{\lambda}{2} = 25 \text{ cm}$$

$$\lambda = 50 \text{ cm}$$

$$v = n\lambda \Rightarrow \sqrt{\frac{T}{\mu}} = n\lambda$$

$$n = 400; T = 20 \text{ N}$$

$$\mu = 0.5 \text{ gm/m}$$

16. **Ans (C)**

$$[V] = [M^0 L^1 T^{-1}]$$

$$[A] = [M^0 L T^{-2}]$$

$$[F] = [M L T^{-2}]$$

$$m \propto V^x A^y F^z$$

$$[M] = [LT^{-1}]^x [LT^{-2}]^y [MLT^{-2}]^z$$

$$M^1 = M^z L^{x+y+z} T^{-x-2y-2z}$$

On comparing the powers we get

$$x = 0; y = -1; z = 1$$

17. **Ans (A)**

$$\Delta X = \lambda$$

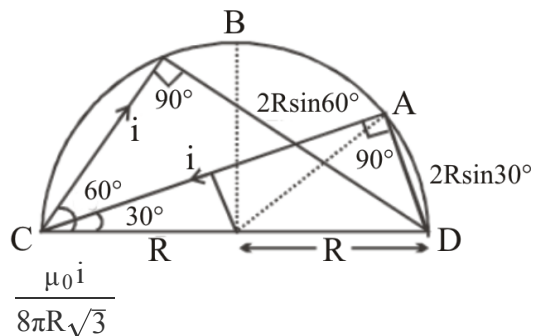
$$\text{So, } I = 4I_0$$

18. **Ans (B)**

$$B_{\text{due to AC}} = \frac{\mu_0 i}{4\pi 2R \sin 30^\circ} [\cos 30^\circ + \cos 90^\circ]$$

$$= \frac{\mu_0 i \sqrt{3}}{8\pi R}$$

$$B_{\text{due to BC}} = \frac{\mu_0 i}{4\pi 2R \sin 60^\circ} [\cos 60^\circ + \cos 90^\circ]$$



$$B_{\text{Net}} = B_{\text{due to AC}} - B_{\text{due to BC}} = \frac{\mu_0 i}{4\pi R \sqrt{3}}$$

19. **Ans (D)**

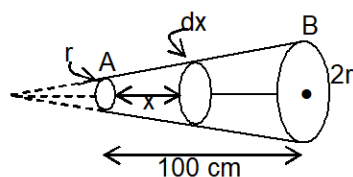
Let the galvanometer be connected at a point $x = x_1$ from end A where $x = 0$.

Let R_1 = resistance of left part i.e. AX_1 and

R_2 = resistance of right part i.e. X_1B

Length = 100 cm = 1 m.

Consider an element of thickness dx at a distance x from end A and of radius r_x



$$\text{Thus, } r_x = \left(r + \frac{r}{1}x\right) = r(1+x)$$

$$\text{Resistance of this element will be, } dR_x = \frac{\rho dx}{\pi r_x^2}$$

$$R_1 = \int_0^{x_1} \frac{\rho dx}{\pi (1+x)^2 r^2} = \frac{\rho}{\pi r^2} \left[1 - \frac{1}{1+x_1}\right]$$

$$R_2 = \int_{x_1}^1 \frac{\rho dx}{\pi (1+x)^2 r^2} = \frac{\rho}{\pi r^2} \left[\frac{1}{1+x_1} - \frac{1}{1+1}\right]$$

For null point or zero deflection,

$$\frac{R_1}{R_2} = \frac{4}{1} \Rightarrow 1 - \frac{1}{1+x_1} = \frac{1}{1+x_1} - \frac{1}{1+1}$$

$$\Rightarrow x_1 = \frac{1}{3} \text{ m} = 33.33 \text{ cm}$$

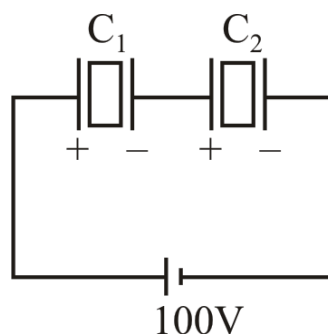
20. **Ans (C)**

$$\left(2 - \frac{6}{8} \times 2\right) = \left(\frac{12}{8+16}\right) 4X \Rightarrow X = 25 \text{ cm}$$

PART-1 : PHYSICS

SECTION-II

1. **Ans (31)**



$$C_1 = \frac{K_1 \epsilon_0 A}{6 \times 10^{-2}} = 50 \epsilon_0 A$$

$$C_2 = \frac{K_2 \epsilon_0 A}{4 \times 10^{-2}} = 50 \epsilon_0 A$$

$$C_{\text{eq}} = 25 \epsilon_0 A$$

$$Q = C_{\text{eq}} \epsilon = 2500 \epsilon_0 A$$

$$Q_{b1} = -Q \left(1 - \frac{1}{K_1}\right), Q_{b2} = Q \left(1 - \frac{1}{K_2}\right)$$

$$Q_b = Q_{b1} + Q_{b2} = Q \left[\frac{1}{K_1} - \frac{1}{K_2}\right] = 2500 \epsilon_0 A \left[\frac{1}{3} - \frac{1}{2}\right]$$

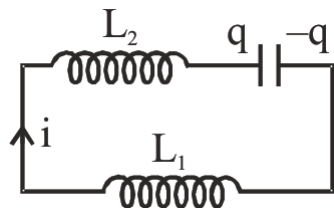
$$Q_b = -2500 \epsilon_0 A \times \frac{1}{6},$$

$$v_b = \frac{Q_b}{A} = -\frac{25}{6} \epsilon_0 A \times 10^2$$

$$M + N = 25 + 6 = 31$$

2. **Ans (6)**

After $t = 0$, circuit will be,



$$-L_1 \frac{di}{dt} - L_2 \frac{di}{dt} - \frac{q}{C} = 0$$

$$\Rightarrow (L_1 + L_2) = \frac{di}{dt} - \frac{q}{C}$$

$$\Rightarrow \frac{d^2q}{dt^2} = -\frac{q}{(L_1 + L_2)C} = -\omega^2 q$$

$$\left\{ \omega = \frac{1}{\sqrt{C(L_1 + L_2)}} \right\}$$

$$\therefore T = \frac{2\pi}{\omega} = 2\pi\sqrt{(L_1 + L_2)C}$$

$$= 2\pi\sqrt{12 \times 10^{-3} \times 30 \times 10^{-6}}$$

$$= 2\pi\sqrt{36 \times 10^{-8}} = 2\pi \times 6 \times 10^{-4} \text{ s}$$

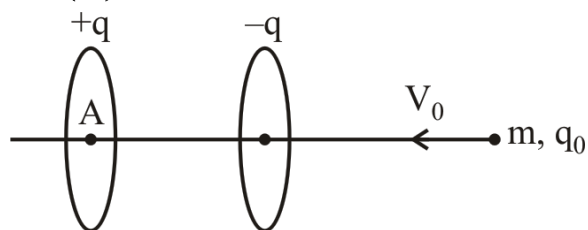
3. **Ans (7)**

$$a = \frac{qE}{m}$$

$$E = \frac{R^2}{2r} \frac{dB}{dt} = \frac{1}{2}(8t - 2) = 7$$

$$x = 7$$

4. **Ans (1)**



Potential energy of q_0 is maximum at point A

for it's path from $+\infty$ to A

by energy conservation

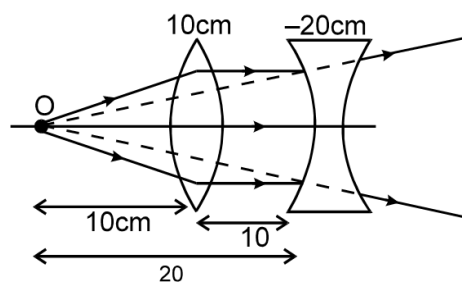
$$\frac{1}{2}mv_0^2 + 0 = 0 + q_0 \left[\frac{kq}{R} - \frac{kq}{2R} \right]$$

$$\Rightarrow V_0 = \sqrt{\frac{kqq_0}{mR}}$$

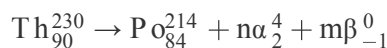
5. **Ans (6)**

$$MP = \left(1 + \frac{D}{f} \right) = \left(1 + \frac{25}{5} \right) = 6$$

6. **Ans (10)**



7. **Ans (2)**



$$230 = 214 + 4n$$

$$n = \frac{16}{4} = 4$$

$$90 = 84 + n \times 2 - m \times 1$$

$$90 = 84 + 4 \times 2 - m \times 1$$

$$m = 92 - 90 = 2$$

$$\text{Hence } \frac{n}{m} = \frac{4}{2} = 2 \text{ Ans.}$$

8. **Ans (5)**

$$f = f_0 \frac{z^2}{n^3} \Rightarrow \frac{z^2}{n^3} = 2 \dots (i)$$

$$E = E_0 \frac{z^2}{n^2} \Rightarrow \frac{z^2}{n^2} = 4 \dots (ii)$$

Solving (i) and (ii) $n = 2, z = 4$

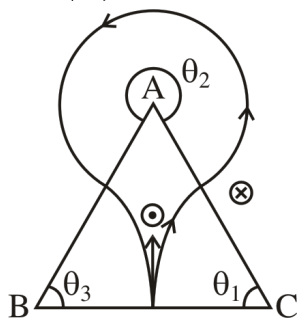
$$L = mvr = \frac{nh}{2\pi}$$

$$\Delta L = \tau \Delta t = \Delta n \frac{h}{2\pi}$$

$$\tau = \frac{\Delta n}{\Delta t} \cdot \frac{h}{2\pi}$$

$$\tau \times 10^{27} = 15$$

9. Ans (7)



Radius of circular path

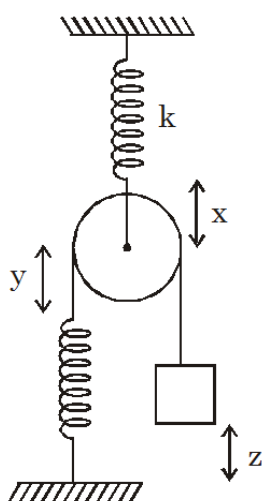
$$R = \frac{mv}{qB} = \frac{0.1 \times \pi}{0.1 \times \pi/3} = 3\text{m}$$

So it will meet side AC at mid-point of AC.
Path of charged particle is shown. So, total angle swept out by charged particle is

$$\theta_1 + \theta_2 + \theta_3 = \frac{7\pi}{3}$$

$$\text{So, time} = \frac{7\pi}{3} \times \frac{m}{qB} = \frac{7\pi}{3} \times \frac{0.1}{0.1 \times \pi/3} = 7\text{sec}$$

10. Ans (5)



$$z = y + 2x$$

$$mgz - \frac{1}{2} \times 4ky^2$$

$$-\frac{1}{2}kx^2 = 0$$

also at any time,

$$kx = 4ky \times 2$$

$$x = 8y$$

$$\Rightarrow z = 17y$$

$$30z - 2k \times \frac{z}{289} - \frac{1}{2}k \times \frac{64z}{289} = 0$$

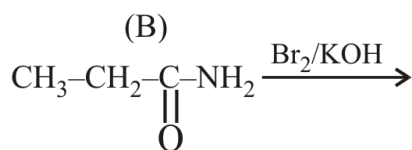
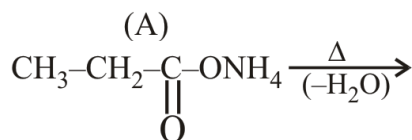
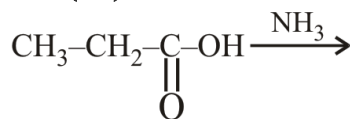
$$15 = \frac{k \times 17z}{289}$$

$$z = \frac{15 \times 17}{5100} = 5\text{cm}$$

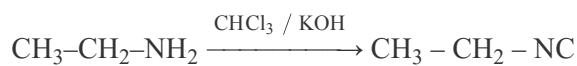
PART-2 : CHEMISTRY

SECTION-I

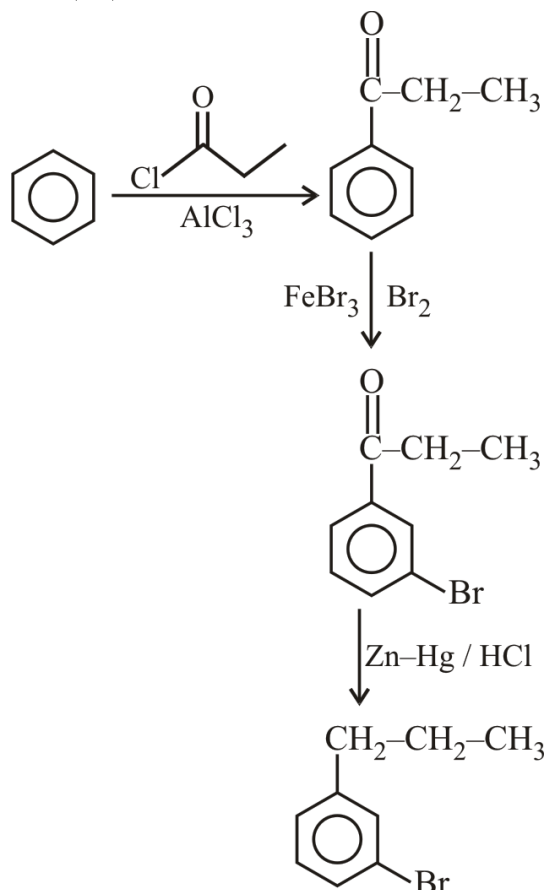
1. Ans (D)



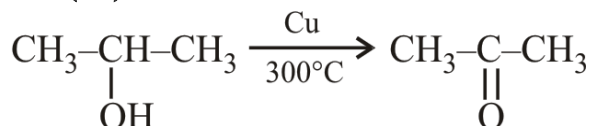
(C)



3. Ans (B)



5. Ans (D)



8. Ans (C)

$$\text{Net Na}^+ = 4 - \left(2 \times \frac{1}{4} + 1\right) = 5/2, \text{Net Cl}^- = 4,$$

$$\text{PF} = \frac{\frac{5}{2} \times \frac{4}{3} \pi r_+^3 + 4 \times \frac{4}{3} \pi r_-^3}{16\sqrt{2}r_-^3}$$

$$\sqrt{2}a = 4r_-$$

$$a = \frac{4r}{\sqrt{2}} = 2\sqrt{2}r_-$$

9. Ans (D)

$$\text{Conductance, } G = \frac{1}{R} = \frac{1}{100} = 0.01 \text{ ohm}^{-1}$$

$$\text{Conductivity, } \kappa = G \cdot G^* = 0.01 \times 0.8$$

$$= 8 \times 10^{-3} \text{ ohm}^{-1} \text{ m}^{-1}.$$

$$\text{Now, } \Lambda_m(\text{HA}) = \frac{\kappa}{C} = \frac{8 \times 10^{-3} \text{ ohm}^{-1} \text{ m}^{-1}}{0.4 \times 10^{-3} \text{ mol m}^{-3}}$$

$$= 2 \times 10^{-5} \text{ ohm}^{-1} \text{ m}^2 \text{ mol}^{-1}$$

$$\alpha = \frac{\Lambda_m}{\Lambda_m^0} = \frac{2 \times 10^{-5}}{0.4 \times 10^{-4}} = 0.5$$

$$\text{Hence, } K_a = \frac{\alpha^2 \cdot C}{1 - \alpha} = \frac{(0.5)^2 \times 0.4}{1 - 0.5} = 0.20$$

10. Ans (A)

More no. of particle, implies more depression in freezing point hence lesser freezing point.

12. Ans (C)

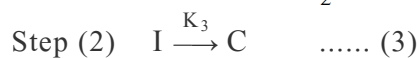
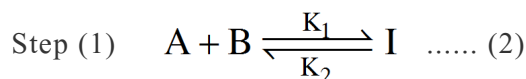
$$W = -P (V_{\text{aragonite}} - V_{\text{calcite}})$$

$$= -\left(100 \times 10^5 \frac{\text{N}}{\text{m}^2}\right) \times \left(\frac{100}{2.5} - \frac{100}{2}\right) \times 10^{-6} \text{ m}^3$$

$$= +100 \text{ J.}$$

13. Ans (C)

For reaction $A + B \rightarrow C$ (1)



Rate for reaction (1)

$$R = K[A]^x [B]^y$$

$$\text{From eq. (2)} \quad \frac{K_1}{K_2} = \frac{[I]}{[A][B]} \quad \text{..... (4)}$$

$$\text{From eq. (3)} \quad R = K_3[I] \quad \text{..... (5)}$$

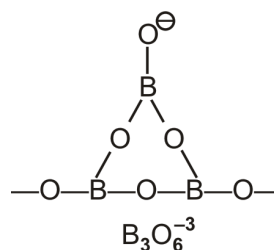
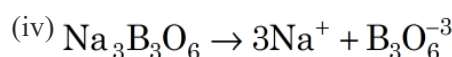
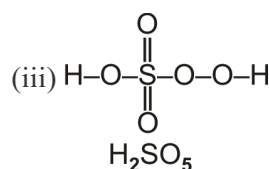
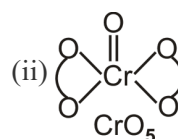
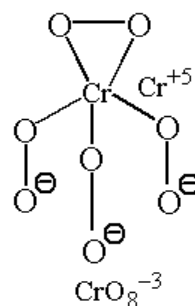
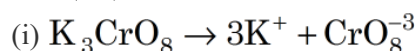
$$\begin{aligned} \text{From eq. (4)} \quad R &= K_3[I] = K_3 \cdot \frac{K_1}{K_2} [A][B] \\ &= K_{\text{overall}} \cdot [A][B] \end{aligned}$$

$$\text{Given : } K_1 = 2 \times 10^3 \text{ L/mole sec}^{-1}$$

$$K_2 = 10^3 \text{ sec}^{-1} \quad K_3 = 10^{-2} \text{ sec}^{-1}$$

$$\begin{aligned} \text{So } K_{\text{overall}} &= \frac{(10^{-2})(\text{sec}^{-1}) \times (2 \times 10^3)(\text{Lmol}^{-1}\text{s}^{-1})}{(10^3)(\text{sec}^{-1})} \\ &= 0.02 \text{ L/mol sec}^{-1} \end{aligned}$$

18. Ans (D)



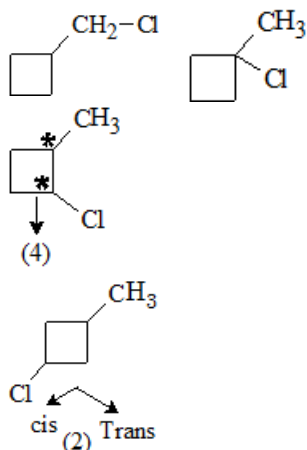
PART-2 : CHEMISTRY

SECTION-II

1. **Ans (5)**

2, 3, 4, 5, 6

2. **Ans (8)**



4. **Ans (1)**

Anode $2\text{H}_2\text{O} \rightarrow \text{O}_2 + 4\text{H}^+ + 4\text{e}^-$

eq. of H^+ produced = $\frac{9.65 \times 100}{96500} = 0.01$

$[\text{H}^+] = 0.1$

pH = 1

5. **Ans (8)**

Coagulating value

$= \frac{2 \times 1}{250} \times 1000 = 8$

6. **Ans (2)**

Formation of buffer solution occurs.

$\text{HCN} + \text{KCN}$
2M, 10mL 5M, V mL

$[\text{HCN}] = \frac{2 \times 10}{10 + V} \text{ M}$

$[\text{KCN}] = \frac{5 \times V}{10 + V}$

$\text{pH} = \text{pK}_a + \log \left(\frac{[\text{KCN}]}{[\text{HCN}]} \right)$

$9 = 10 - \log 5 + \log \left(\frac{5V}{20} \right)$

$\log 5 - 1 = \log \left(\frac{5V}{20} \right)$

$\log \left(\frac{5}{10} \right) = \log \left(\frac{5V}{20} \right)$

$\frac{1}{2} = \frac{5V}{20}$

$V = 2 \text{ mL}$

7. **Ans (3)**

$n_{\text{eq}} \text{N}_2\text{H}_4 = n_{\text{eq}} \text{K}_2\text{CrO}_4$

or, $\frac{w}{32} \times 4 = \frac{24.25}{194} \times 3 \Rightarrow w = 3 \text{ gm}$

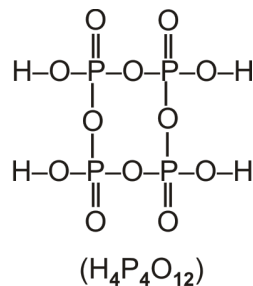
8. **Ans (4448)**

a = 4

b = 4

c = 4

d = 8



10. **Ans (6619)**

For d^4 ion (Cr^{2+} ion) in HS state,

CFSE = $-0.6\Delta_0$

$= -0.6 \times 13900 \text{ cm}^{-1}$

$= -8340 \text{ cm}^{-1}$ (Answer)

For d^4 ion (Cr^{2+} ion) in LS state,

CFSE = $-1.6\Delta_0 + P$

$= (-1.6 \times 13900 + 23500) \text{ cm}^{-1}$

$= +1260 \text{ cm}^{-1}$ (Answer)

Since C.F.S.E. in HS state is less so more stable

$\frac{x}{y} = \left(\frac{8340}{1260} \right) \times 1000$
 $= 6619$

PART-3 : MATHEMATICS

SECTION-I

1. **Ans (C)**

Let ED be the pole and EC the flagstaff. A be the nearer point. $\text{AD} = \text{DC} = 4 \Rightarrow \angle \text{CAD} = 45^\circ$
so solve $\triangle \text{EAD}$.

2. **Ans (C)**

Point of intersection is (2, 1). Slope of tangents are $1/2$ and -2 respectively.

3. **Ans (D)**

$$\sqrt{2}(\cos 3x - \sin 3x) = \sqrt{3} \sin 2x + \cos 2x$$

$$\frac{(\cos 3x - \sin 3x)}{\sqrt{2}} = \left(\frac{\sqrt{3} \sin 2x + \cos 2x}{2} \right)$$

$$\cos\left(3x + \frac{\pi}{4}\right) = \cos\left(2x - \frac{\pi}{3}\right)$$

$$3x + \frac{\pi}{4} = 2n\pi \pm \left(2x - \frac{\pi}{3}\right)$$

$$x = 2n\pi - \frac{7\pi}{12}, x = \frac{2n\pi}{5} + \frac{\pi}{60}; n \in \mathbb{Z}$$

4. **Ans (B)**

Domain $-1 \leq x \leq 1$. From graph of functions $y = \cot^{-1} x$ and $y = \sin^{-1} x$, solution must be of the form $[-1, \alpha]$ where $0 < \alpha < 1$ satisfies $\cot^{-1} \alpha = \sin^{-1} \alpha$. Take $\sin$ on both sides, put $\cot^{-1} \alpha = \theta$ and solve using right angled triangle.

5. **Ans (C)**

$$\text{Put } \int_0^{\pi/2} \theta \sin^8 \theta d\theta = I, \int_0^{\pi/2} \theta \sin^{10} \theta d\theta = P$$

$$P = \int_0^{\pi/2} (1 - \cos^2 \theta) \sin^8 \theta d\theta = I, - \int_0^{\pi/2} \theta \cos^2 \theta (\sin^8 \theta) d\theta$$

$$= I - \int_0^{\pi/2} (\theta \cos \theta)(\cos \theta \sin^8 \theta) d\theta. \text{ Using integration by parts we get}$$

$$P = I - \left[\frac{(\theta \cos \theta) \sin^9 \theta}{9} - \int \left(\frac{-\theta \sin \theta + \cos \theta}{9} \right) \sin^9 \theta d\theta \right]_0^{\pi/2}$$

$$P = I - \left(\frac{P}{9} \right) + \frac{1}{9} \int_0^{\pi/2} \sin^9 \theta \cos \theta d\theta, \frac{10P}{9} = I + \frac{1}{90}$$

$$\therefore 100P = 90I + 1$$

$$\therefore \frac{100P - 1}{I} = 90$$

6. **Ans (A)**

$$\text{Differentiate to get } y' = \frac{-9(x - \sqrt{1+x^2})^9}{\sqrt{1+x^2}}$$

$$\text{Rewrite as } \sqrt{1+x^2} \cdot y' = -9y \dots (1)$$

$$\text{Differentiate (1) both sides w.r.t. } x, \text{ then put value of } \sqrt{1+x^2} \cdot y' \text{ from (1).}$$

7. **Ans (C)**

p means 'at least one prime number is even'. So its negation is 'No prime number is even'.

8. **Ans (A)**

Let x be thickness of ice at time t .
So $\frac{d}{dt} \left(\frac{4}{3} \pi (20+x)^3 - \frac{4}{3} \pi (20)^3 \right) = -50$.
Use chain rule and put $x = 5$.

9. **Ans (D)**

By inspection $f(\pi/2) = 3\pi$, as $g(x) = f^{-1}(x)$, $g(f(x)) = x$.
Taking derivative we get $f'(x) g'(f(x)) = 1 \dots (i)$
Taking derivative again we get $f''(x) g'(f(x)) + f'(x)^2 g''(f(x)) = 0 \dots (ii)$
 $f(x) = 4x + \pi \sin x \dots (iii)$
From (iii) we can get the value of $f'(\pi/2)$ and $f''(\pi/2)$.
By substituting $x = \pi/2$ in (i) and (ii) we can get the value of $g'(3\pi)$ and $g''(3\pi)$.

10. **Ans (A)**

$g(x) = 12x^5 + 15ax^4 + 20bx^3 + 30cx^2 + 60dx + k = 0$
has 5 positive real roots.
 $\therefore$ By Rolle's theorem $g'(x) = 0$ has four positive real roots.
 $\therefore f(x) = 0$ has four positive real roots $\alpha, \beta, \gamma, \delta$ such that
$$\frac{\alpha + \beta + \gamma + \delta}{4} = \frac{4}{(1/\alpha) + (1/\beta) + (1/\gamma) + (1/\delta)}$$

 $\Rightarrow AM = HM \Rightarrow \alpha = \beta = \gamma = \delta$
 $\therefore f(x) = (x - \alpha)^4 \Rightarrow f(2) = (2 - \alpha)^4 = 1$
 $\Rightarrow 2 - \alpha = 1 \text{ or } \Rightarrow \alpha = 1 \text{ or } 3$.

11. **Ans (D)**

Domain is $\mathbb{R}$. Clearly we need to check differentiability only at $x = 0$. Use first principle.

12. **Ans (C)**

Centre $(2, k)$ and passing through $(0, 0)$ so
 $(x - 2)^2 + (y - k)^2 = 2^2 + k^2$
 $\Rightarrow x^2 + y^2 - 4x - 2ky = 0 \dots (1)$ differentiate and eliminate k .

13. **Ans (C)**

H : hit, X : not hit. $P(HXXXH \text{ or } XHXXH \text{ or } XXHH) = 3 \times (2/3)^2 \times (1/3)^2$.

14. **Ans (C)**

Put $x - \pi/4 = t$, expand numerator and integrate.
OR check options by differentiating.

15. **Ans (D)**

$$AD = 4$$

$$AP = \frac{2}{3} \times 4 = \frac{8}{3}$$

In $\triangle ABP$

$$\frac{BP}{AP} = \tan \frac{\pi}{6} \Rightarrow BP = \frac{8}{3\sqrt{3}}$$

$$\text{Area of } \triangle ABC = 3 \times \text{Area of } \triangle ABP = \frac{32}{3\sqrt{3}}$$

16. **Ans (D)**

$$a^2 + (b - c)^2 = 0$$

$$\Rightarrow a = 0 \text{ and } b = c \neq 0 \quad \because a^2 + b^2 + c^2 \neq 0$$

$\therefore$ Required line becomes $by + c = 0$

$$\therefore y = -1$$

$\therefore (2, -1)$ lies on line

17. **Ans (C)**

$$\text{Let } z = x + iy$$

$$-8((x + iy)^2 + (x - iy)^2) + 20(x^2 + y^2) = 144$$

$$-8(2(x^2 - y^2)) + 20(x^2 + y^2) = 144$$

$$4x^2 + 36y^2 = 144$$

$$\frac{x^2}{36} + \frac{y^2}{4} = 1$$

$$a = 6, b = 2$$

$$\text{Area} = \pi ab = 12\pi$$

18. **Ans (D)**

Let n = number of trials

p = chance of success

q = chance of failure

$$np = 4 \quad npq = \frac{4}{3}$$

$$q = \frac{1}{3} \quad p = \frac{2}{3} \quad n = 6$$

let x = number of success

$$p(x \geq 2) = 1 - p(x = 0) - p(x = 1)$$

$$= 1 - {}^6C_0 q^6 - {}^6C_1 \cdot p q^5 = \frac{716}{729}$$

19. **Ans (C)**

$$\text{Put } 5^x = t$$

$$t^2 + t - m < 0$$

For at least one $t \in \mathbb{R}^+$

$$\text{Since } -\frac{b}{2a} < 0$$

$$\Rightarrow f(0) < 0$$

$$-m < 0$$

$$\Rightarrow m > 0$$

20. **Ans (D)**

Let roots are $\frac{k}{r}, k, kr$

$$\Rightarrow k = -6$$

$$\Rightarrow \frac{k}{r} + k + kr = -a \quad \dots(1)$$

$$\frac{k^2}{r} + k^2 r + k^2 = b \quad \dots(2)$$

Solving (1) & (2) we set $\frac{b}{a} = 6$

$$\frac{a^2 + b^2}{ab} \Rightarrow \frac{a}{b} + \frac{b}{a}$$

$$= 6 + \frac{1}{6}$$

$$= \frac{37}{6}$$

PART-3 : MATHEMATICS

SECTION-II

1. **Ans (6)**

Solve first two equations to get $x = 2 + z$ and

$y = 4 - 2z$. Put in third equation to get $(\lambda - 6)z = 4$.

No solution for $\lambda = 6$.

$$\text{OR } \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 2 & 4 & \lambda \end{vmatrix} = 0 \Rightarrow \lambda = 6.$$

2. **Ans (0)**

$A^2B + AB^2 = 0$. Pre-multiply both sides by A to get $A^3B + A^2B^2 = (-B^3)B + A^2B^2 = (-B^2 + A^2)B^2 = 0$.

Take determinant on both sides to get $|A^2 - B^2| = 0$ or $|B| = 0$. If $|B| = 0$ then taking determinant on both sides of $A^3 = -B^3$ gives $|A| = 0$, which is contradiction.

3. **Ans (1)**

Multiply and divide denominator by x^2 to remove $\sin^2 x$ then use L' Hospital rule (Use Leibniz rule for differentiation of integral)

4. **Ans (20)**

Let number of elements in S be k . $10k = 5n$ and $9k = 90$.

5. **Ans (4)**

$AB = I$. So $(1, 1)^{\text{th}}$ element of AB must be 1.

6. **Ans (624)**

$2520 = 2^3 \cdot 3^2 \cdot 5 \cdot 7$ Required sum is $(3^0 + 3^1 + 3^2) (5^0 + 5^1) (7^0 + 7^1)$.

7. **Ans (11)**

Put $z = \omega$ or $z = \omega^2$.

8. **Ans (2)**

Direct method : area = $\int_{(1-e)/2}^0 \log_e (2x + e) dx$.

Now use integration by parts. Faster method

$$: \text{area} = \int_0^1 x \, dy = -(1/2) \int_0^1 (e^y - e) dy$$

9. **Ans (2)**

Replace x by $a + b - x$ then add the two integrals

10. **Ans (2)**

Circle $(x - r)^2 + (y - r)^2 = r^2$

(x_1, y_1) lies on it $(x_1 - r)^2 + (y_1 - r)^2 = r^2$

$$r^2 - 2r(x_1 + y_1) + x_1^2 + y_1^2 = 0$$

$$x_1^2 + y_1^2 = 2r(x_1 + y_1)$$

$$\frac{x_1^2 + y_1^2}{x_1 + y_1} = 2$$

$$\frac{(x_1 + y_1)^2 - 2x_1 y_1}{(x_1 + y_1)} = 2$$